

ANALYTICAL PROBLEM-SOLVING ACTIVITY

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Topic 1: Metrics for Performance Evaluation

1. Accuracy

A classification model tested 100 samples. It correctly classified 85 samples.

Q1. How many samples were classified incorrectly?

Ans: Incorrect samples = Total samples – Correctly classified samples = $100 - 85 = 15$ samples.

Q2. Calculate the accuracy of the model.

Ans: Accuracy = (Correctly classified samples / Total samples) $\times 100 = (85/100) \times 100 = 85\%$.

Q3. Calculate the error rate.

Ans: Error Rate = $100\% - \text{Accuracy} = 100\% - 85\% = 15\%$ (or $\text{Incorrect/Total} \times 100 = 15/100 \times 100 = 15\%$).

Q4. Is the model performing well based on accuracy?

Ans: Yes, an accuracy of 85% is generally considered good performance, though this should be confirmed with other metrics (precision, recall) especially if the dataset is imbalanced.

Q5. What happens to accuracy if 5 more samples are classified correctly?

Ans: New correctly classified samples = $85 + 5 = 90$. New Accuracy = $(90/100) \times 100 = 90\%$. So accuracy increases from 85% to 90%.

2. Confusion Matrix

$TP = 40, TN = 35, FP = 10, FN = 15$

Q1. What is the total number of predictions?

Ans: Total = $TP + TN + FP + FN = 40 + 35 + 10 + 15 = 100$ predictions.

Q2. Calculate the accuracy.

Ans: Accuracy = $(TP + TN) / \text{Total} \times 100 = (40 + 35)/100 \times 100 = 75\%$.

Q3. Calculate the error rate.

Ans: Error Rate = $100\% - \text{Accuracy} = 100\% - 75\% = 25\%$ (or $(FP + FN)/\text{Total} \times 100 = 25/100 \times 100 = 25\%$).

Q4. Which value represents incorrectly predicted positive cases?

Ans: False Positive (FP) = 10 represents cases incorrectly predicted as positive (actually negative).

Q5. Which value represents incorrectly predicted negative cases?

Ans: False Negative (FN) = 15 represents cases incorrectly predicted as negative (actually positive).

3. Precision and Recall

A disease prediction system: TP = 80, FP = 20, FN = 10, TN = 90

Q1. Calculate Precision.

Ans: Precision = $TP / (TP + FP) = 80 / (80 + 20) = 80/100 = 0.80 = 80\%$.

Q2. Calculate Recall.

Ans: Recall = $TP / (TP + FN) = 80 / (80 + 10) = 80/90 \approx 0.889 = 88.89\%$.

Q3. Which metric tells us how many predicted positive cases were actually positive?

Ans: Precision tells us this — it measures the correctness of positive predictions.

Q4. Which metric tells us how many actual positive cases were correctly identified?

Ans: Recall (Sensitivity) tells us this — it measures how many actual positives were captured by the model.

Q5. Which metric is more important when missing a positive case is dangerous?

Ans: Recall is more important in this situation (e.g., disease detection), because a False Negative (missing an actual positive/diseased case) can have serious consequences.

4. Compare Two Classifiers

Two classifiers are tested on the same dataset:

Metric	Classifier A	Classifier B
Accuracy	90%	85%
Precision	80%	95%
Recall	70%	90%

Q1. Which classifier has higher accuracy?

Ans: Classifier A has higher accuracy (90% vs 85%).

Q2. Which classifier has higher precision?

Ans: Classifier B has higher precision (95% vs 80%).

Q3. Which classifier has higher recall?

Ans: Classifier B has higher recall (90% vs 70%).

Q4. Which classifier would you choose if detecting as many positive cases as possible is important?

Ans: Classifier B, because it has a higher recall (90%), meaning it correctly identifies more of the actual positive cases.

Q5. Give one reason why accuracy alone may not be enough.

Ans: Accuracy can be misleading on imbalanced datasets — a model can show high accuracy while still performing poorly at identifying the minority (positive) class, which is why precision and recall are also needed.

5. Analyze Model Performance

A classification model produces: TP = 70, TN = 80, FP = 20, FN = 30

Q1. Calculate accuracy.

Ans: Accuracy = $(TP + TN) / \text{Total} = (70 + 80) / (70+80+20+30) = 150/200 = 0.75 = 75\%$.

Q2. Calculate precision.

Ans: Precision = $TP / (TP + FP) = 70 / (70 + 20) = 70/90 \approx 0.778 = 77.78\%$.

Q3. Calculate recall.

Ans: Recall = $TP / (TP + FN) = 70 / (70 + 30) = 70/100 = 0.70 = 70\%$.

Q4. Calculate the error rate.

Ans: Error Rate = $100\% - \text{Accuracy} = 100\% - 75\% = 25\%$.

Q5. Based on these values, identify one strength and one weakness of the model.

Ans: Strength: Reasonably high precision (~77.78%) — when it predicts positive, it is usually correct.
Weakness: Recall (70%) is comparatively lower, meaning the model misses a noticeable number (30) of actual positive cases (FN = 30).

Topic 2: Prediction Algorithms

1. Simple Linear Prediction

A company's sales are predicted based on advertising expenditure (linear relationship).

Advertising Cost (₹)	Sales
10	20
20	40
30	60
40	80

Q1. Identify the relationship between advertising cost and sales.

Ans: Sales = $2 \times \text{Advertising Cost}$. Sales is directly proportional to advertising cost, growing at a constant rate — a linear relationship ($y = 2x$).

Q2. Predict sales when advertising cost is ₹50.

Ans: Sales = $2 \times 50 = ₹100$.

Q3. What happens to sales when advertising cost increases by ₹10?

Ans: Sales increases by ₹20 (since Sales = $2 \times \text{Cost}$, a ₹10 increase in cost produces a ₹20 increase in sales).

Q4. Is this an example of classification or prediction?

Ans: This is an example of (numerical) prediction, since the output is a continuous numeric value (sales amount), not a category/class label.

Q5. Why is a linear prediction algorithm suitable here?

Ans: Because the data shows a constant, proportional rate of change between advertising cost and sales — the points fit a straight line, so linear regression can model this relationship accurately.

2. Predict Temperature

A simple prediction model is: Temperature ($^{\circ}\text{C}$) = $25 + 1 \times (\text{Day number})$. [Note: the source document did not include the explicit formula/data table for this question, so a representative linear model with a constant daily

increase of 1°C and a base temperature of 25°C on Day 0 is assumed for illustration, consistent with the linear-prediction pattern used across this activity.]

Q1. Predict the temperature on Day 5.

Ans: Temperature = $25 + 1 \times 5 = 30^{\circ}\text{C}$.

Q2. Predict the temperature on Day 10.

Ans: Temperature = $25 + 1 \times 10 = 35^{\circ}\text{C}$.

Q3. What is the predicted temperature on Day 15?

Ans: Temperature = $25 + 1 \times 15 = 40^{\circ}\text{C}$.

Q4. How much does the temperature increase for every additional day?

Ans: The temperature increases by 1°C for every additional day (this is the slope of the linear model).

Q5. Is this a classification or prediction problem?

Ans: This is a (numerical) prediction problem, since the model outputs a continuous numeric value (temperature) rather than a class label.

3. House Price Prediction

A prediction algorithm estimates house prices based on area (linear relationship).

Area (sq.ft)	Price (₹ lakhs)
1000	40
1500	60
2000	80

Q1. Identify the relationship between area and price.

Ans: Price = $0.04 \times \text{Area}$. The price increases proportionally with area — a linear relationship (₹4,000 per sq.ft, i.e., 0.04 lakh per sq.ft).

Q2. Predict the price of a 2500 sq.ft house.

Ans: Price = $0.04 \times 2500 = ₹100$ lakhs.

Q3. What happens to the price when area increases by 500 sq.ft?

Ans: Price increases by $0.04 \times 500 = ₹20$ lakhs.

Q4. Which type of algorithm can be used for this problem?

Ans: A Linear Regression algorithm can be used, since it models the straight-line relationship between a numeric input (area) and a numeric output (price).

Q5. Is this classification or numerical prediction?

Ans: This is numerical prediction (regression), because the output is a continuous value (price in lakhs), not a discrete class.

4. Compare Actual and Predicted Values

A prediction algorithm gives the following results:

Actual Value	Predicted Value	Error = Actual – Predicted
50	48	2
60	65	5

70	68	2
80	85	5
90	88	2

Q1. Which prediction has the smallest error?

Ans: The predictions for Actual = 50 (Error = 2), Actual = 70 (Error = 2), and Actual = 90 (Error = 2) all tie for the smallest error, each with an error of 2.

Q2. Which prediction has the largest error?

Ans: The predictions for Actual = 60 (Predicted = 65) and Actual = 80 (Predicted = 85) tie for the largest error, each with an error of 5.

Q3. Calculate the error for each prediction.

Ans: 50→48: $|50-48| = 2$ | 60→65: $|60-65| = 5$ | 70→68: $|70-68| = 2$ | 80→85: $|80-85| = 5$ | 90→88: $|90-88| = 2$.

Q4. Is the model generally predicting close to the actual values?

Ans: Yes, the model is predicting reasonably close to actual values — errors range only from 2 to 5, which is small relative to the magnitude of the values (50–90), giving an average error of about 3.2.

Q5. Why is the difference between actual and predicted values important?

Ans: This difference (the error/residual) tells us how accurate the prediction model is. Smaller errors indicate a better-fitting model, and analyzing the pattern of errors (e.g., using metrics like MAE or RMSE) helps evaluate and improve model performance.

5. Choosing a Prediction Algorithm

Consider the following applications: A. Predicting tomorrow's temperature. B. Predicting the price of a house. C. Predicting the number of products sold next month. D. Predicting whether an email is spam or not.

Q1. Which applications are prediction problems?

Ans: A (temperature), B (house price), and C (number of products sold) are prediction (regression) problems — each produces a continuous numeric output.

Q2. Which application is a classification problem?

Ans: D — predicting whether an email is spam or not — is a classification problem, since the output is a discrete category (Spam / Not Spam).

Q3. Which algorithm could be used for predicting house prices?

Ans: Linear Regression (or other regression algorithms such as Decision Tree Regression, Random Forest Regression) can be used for predicting house prices.

Q4. Why is regression suitable for numerical prediction?

Ans: Regression models the relationship between input variables and a continuous output variable, allowing it to estimate numeric values (like price or temperature) rather than assigning discrete class labels.

Q5. Give one difference between classification and prediction.

Ans: Classification predicts a discrete category/class label (e.g., spam or not spam), whereas prediction (regression) estimates a continuous numeric value (e.g., price, temperature, sales).